

Algebraic Reduction of the Syracuse Inverse Triangle to Its Leftmost Branch

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Abstract. We study the inverse geometry of the Syracuse map through the triangle construction arising in the measure-theoretic framework of Assani, Ebbighausen, and Hande. The project materials isolate a distinguished family of nodes of the form $3^a 2^b h - 1$, with h coprime to 6 , and describe two inverse actions: left division $L(x) = (2x - 1)/3$ and right doubling $R(x) = 2x$.

The main contribution of this paper is an explicit algebraic reduction for every *pure left-action* branch: we prove the closed form

$$L^i(3^k h - 1) = 3^{k-i} 2^i h - 1,$$

and for every $1 \leq i < k$, this value is strictly smaller than the generating node $3^k h - 1$. We also clarify the exact length of pure-left chains as $v_3(r + 1)$ (number of steps) or $1 + v_3(r + 1)$ (number of nodes).

Crucially, we demonstrate that the unresolved part of the triangle claim is localized precisely to the leftmost boundary, where the inequality fails and the trajectory folds back into itself. We emphasize that this reduction does **not** prove the Collatz conjecture; the leftmost branch remains the essential obstruction, and finite verification cannot substitute for a global proof.

1. Introduction

The Collatz problem, one of the most famous unsolved problems in number theory, asks whether every positive integer eventually reaches the trivial cycle $1 \leftrightarrow 2$ under the map $n \mapsto n/2$ if even, and $n \mapsto 3n + 1$ if odd. The Syracuse formulation compresses all divisions by 2 after an odd step into a single map:

$$T(n) = \begin{cases} n/2, & n \text{ even,} \\ (3n + 1)/2^{v_2(3n+1)}, & n \text{ odd.} \end{cases}$$

Inverse-map analysis has attracted considerable attention because the forward conjecture can be recast as a question about the geometry and growth of preimage trees.

Assani and collaborators developed a measure-theoretic framework in which inverse-image structure plays a central role. Their work proves, among other results, an equivalence between the existence of a suitable finite power-bounded measure and the global cycle property [1, 2]. The present paper does not reproduce that theory. Instead, it isolates a concrete algebraic feature of the inverse triangles and turns a broad empirical claim into a sharply localized obstruction.

2. Preliminaries: Syracuse Map and Inverse Actions

Let $v_2(m)$ denote the exponent of the highest power of 2 dividing m . The Syracuse map is defined as above.

For the relevant N_2 -type nodes appearing in the inverse tree, the project materials use the representation:

$$n = 3^a 2^b h - 1,$$

where $a > 0$, $b \geq 0$, and h is not divisible by 2 or 3 (i.e., $\gcd(h, 6) = 1$).

The inverse dynamics are generated by two branches:

- **Left inverse branch** (when defined as an integer): $L(x) = \frac{2x-1}{3}$. For $x = 3^a 2^b h - 1$ with $a \geq 1$, we have: $L(x) = 3^{a-1} 2^{b+1} h - 1$.
- **Right inverse branch** (always defined): $R(x) = 2x$.

The triangle generated by a root $r = 3^k h - 1$ consists of all nodes obtainable by applying sequences of L and R of total length up to k , provided each L -application remains within the integer domain.

3. Main Algebraic Lemma

We first establish the exact algebraic form of repeated left actions.

Lemma 1 (Pure Left Iteration).

For integers $k \geq 0$, h coprime to 6, and $0 \leq i \leq k$, the repeated left action satisfies:

$$L^i(3^k h - 1) = 3^{k-i} 2^i h - 1.$$

Proof. We proceed by induction on i . For $i = 0$, the identity is trivial. Assume it holds for some $i < k$. Then,

$$\begin{aligned}
L^{i+1}(3^k h - 1) &= L(L^i(3^k h - 1)) \\
&= L(3^{k-i} 2^i h - 1) \\
&= \frac{2(3^{k-i} 2^i h - 1) - 1}{3} \\
&= \frac{3^{k-i} 2^{i+1} h - 3}{3} \\
&= 3^{k-i-1} 2^{i+1} h - 1.
\end{aligned}$$

Thus the formula holds by induction. $\square$

Remark 1 (Domain Boundary).

For $i = k$, the formula gives $L^k(3^k h - 1) = 2^k h - 1$. At this boundary point, the exponent of 3 is zero, so L is not defined as an integer (since $3 \nmid (2(2^k h - 1) - 1)$). Therefore, Lemma 1 is algebraically valid as a boundary value, but the inverse action L cannot be applied further. The "leftmost branch" consists precisely of these boundary nodes.

4. Strict Descent Away from the Left Boundary

Theorem 2 (Strict Descent for Non-Boundary Branches).

Let $r = 3^k h - 1$ with $k \geq 1$ and $\gcd(h, 6) = 1$. For every $1 \leq i < k$,

$$L^i(r) = 3^{k-i} 2^i h - 1 < r = 3^k h - 1.$$

Proof. By Lemma 1, it suffices to compare:

$$3^{k-i} 2^i h < 3^k h.$$

Since $3^{k-i} h > 0$, we cancel it to obtain:

$$2^i < 3^i,$$

which is true for every $i \geq 1$. Hence the inequality holds strictly. $\square$

Corollary 1 (Reduction of the Triangle Claim).

Except for the leftmost boundary nodes $2^i h - 1$ (where the exponent of 3 is zero), every pure left-descendant of a root is strictly smaller than the root.

5. Why the Leftmost Branch Is Exceptional

The project materials record the boundary behavior:

$$L^k(3^k h - 1) = 2^k h - 1.$$

For odd h , this boundary point lies outside the finite triangle layer. Applying the forward Syracuse map T to this boundary node yields:

$$T(2^k h - 1) = 3 \cdot 2^{k-1} h - 1,$$

which is precisely the preceding left-branch node at depth $k - 1$.

This "folding back" explains structurally why the non-left branches can be handled by Theorem 2 while the leftmost branch cannot simply be treated by the same inequality. The boundary trajectory does not produce a strict numerical descent immediately; instead, it returns to the left branch, potentially allowing for infinite cycles.

6. Exact Pure-Left Chain Length

Let r be a root. The repeated left action changes $r + 1$ according to:

$$L(m) + 1 = \frac{2(m + 1)}{3}.$$

Thus, one left step decreases $v_3(m + 1)$ by exactly one. Consequently:

- The number of **available left steps** is $v_3(r + 1)$.
- The number of **nodes in the chain** is $1 + v_3(r + 1)$.

For $r = 3^k h - 1$, we have $v_3(r + 1) = k$, consistently with Lemma 1.

7. Relation to the Assani Triangle Claim

The broader claim states: for every point in a triangle, except the left branch, there exists an iterate strictly smaller than the starting point. The present theorem proves the central algebraic part for left-action descendants. We formally state the reduction:

$$\text{Full Triangle Descent Claim} \iff \text{Leftmost Boundary Problem.}$$

8. Significance and Limitations

Significance. The main value is reduction rather than completion. A proof of the remaining boundary mechanism would have a direct route back into the inverse-tree program.

Limitations. The theorem does not prove the Collatz conjecture. Finite verification cannot establish the global claim. The leftmost branch remains unresolved.

9. Conclusion

We obtain an explicit reduction:

$$L^i(3^k h - 1) = 3^{k-i} 2^i h - 1 < 3^k h - 1 \quad (1 \leq i < k).$$

Together with the exact pure-left chain length and the boundary folding analysis, this provides an auditable algebraic core for the inverse-triangle program. The next target is a proof that the leftmost boundary cannot sustain an infinite non-descending mechanism.

References

1. I. Assani, "Collatz map as a non-singular transformation," *arXiv:2208.11675*, 2022.
2. I. Assani, E. Ebbighausen, and A. Hande, "Syracuse Maps as Non-singular Power-Bounded Transformations and Their Inverse Maps," *arXiv:2208.11801*, 2022.
3. Project Source Materials: *collatz_exploration.md*, *derive.py*, *verify2.py* (internal).